Euler's Method and Logistic Growth

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = a$ and a local minimum at $x = b$. The tangent line to the graph of f at $x = c$ has the equation $y - f(c) = \frac{1}{2}(x - c)$. Given that $f(c) = 3$ and $f'(c) = \frac{1}{2}$, find the value of c.

The equation of the tangent line to the graph of a function f at $x = c$ is given by $y - f(c) = f'(c)(x - c)$. We are given that the equation of the tangent line at $x = c$ is $y - f(c) = \frac{1}{2}(x - c)$. Comparing this to the general form, we see that $f'(c) = \frac{1}{2}$. We are also given that $f(c) = 3$. The graph shows a point on the function $f(x)$ at $(x, y) = (2, 3)$. Since the point $(c, f(c))$ must lie on the graph of f, we have $c = 2$ and $f(c) = 3$. Therefore, the value of c is 2.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A particle's position in the xy-plane is given by the parametric equations $x(t) = at^2 - 3t$ and $y(t) = 2t^3 + bt$, where a and b are constants. At time $t = 1$, the particle's velocity vector is $\mathbf{v}(1) = \langle 9, 10 \rangle$. Find the value of a and the value of b.

The velocity vector is given by the derivatives of the position components with respect to time: $x'(t) = \frac{dx}{dt} = 2at - 3$ and $y'(t) = \frac{dy}{dt} = 6t^2 + b$. We are given that at $t = 1$, the velocity vector is $\mathbf{v}(1) = \langle 9, 10 \rangle$. Therefore, we can set up the following system of equations: For the x-component: $x'(1) = 2a(1) - 3 = 9\ 2a - 3 = 9\ 2a = 12\ a = 6$ For the y-component: $y'(1) = 6(1)^2 + b = 10$ $6 + b = 10\ b = 4$ Thus, the values are $a = 6$ and $b = 4$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Give the first three nonzero terms of the Maclaurin series for $\sin x$.

Theorem (Maclaurin Series for Sine): The sine function is odd, so its Maclaurin series contains exclusively odd powers of x with alternating signs. Step 1 · Compute successive derivatives at $x = 0$: $f(x) = \sin x \Rightarrow f(0) = 0\ f'(x) = \cos x \Rightarrow f'(0) = 1\ f''(x) = -\sin x \Rightarrow f''(0) = 0\ f'''(x) = -\cos x \Rightarrow f'''(0) = -1$ $f^{(4)}(x) = \sin x \Rightarrow f^{(4)}(0) = 0\ f^{(5)}(x) = \cos x \Rightarrow f^{(5)}(0) = 1$ Step 2 · Form the first three nonzero terms.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Give the first three terms of the Maclaurin series for $\cos x$.

Theorem (Maclaurin Series for Cosine): The cosine function is even, so its Maclaurin series consists exclusively of even powers of x with alternating signs. Step 1 · General Maclaurin formula for $\cos x$: $\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$. Step 2 · Write out the terms for $n = 0, 1, 2$: $n = 0 \Rightarrow (-1)^0 \frac{x^0}{0!} = 1$ $n = 1 \Rightarrow (-1)^1 \frac{x^2}{2!} = -\frac{x^2}{2}$ $n = 2 \Rightarrow (-1)^2 \frac{x^4}{4!} = +\frac{x^4}{24}$ Conclusion: The first three terms are.

5 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x) = x^2 - 6x + 10$. Find the x-coordinates of the points on the graph of $y = f(x)$ where the tangent line is horizontal.

A horizontal tangent line occurs where the derivative of the function is equal to zero. First, find the derivative of $f(x)$: $f'(x) = \frac{d}{dx}(x^2 - 6x + 10) = 2x - 6$. Next, set the derivative equal to zero and solve for x : $2x - 6 = 0$. Adding 6 to both sides gives $2x = 6$. Dividing by 2 gives $x = 3$. Therefore, the x-coordinate of the point where the tangent line is horizontal is $x = 3$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For what x does $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converge?

Theorem (Ratio Test for Absolute Convergence): For an infinite series $\sum a_n$, evaluate $L = \lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|$. The series converges absolutely when $L < 1$. Step 1 · Set up the ratio for $a_n = \frac{x^n}{n!}$: $L = \lim_{n \to \infty} \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \lim_{n \to \infty} \left| \frac{x \cdot x^n}{(n+1) \cdot n!} \cdot \frac{n!}{x^n} \right| = \lim_{n \to \infty} \frac{|x|}{n+1}$. Step 2 · Evaluate the limit f.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The polar curve $r = 2\cos\theta$ is a circle. Find its enclosed area.

Theorem (Area of Circle in Polar Coordinates): The curve $r = 2\cos\theta$ is a circle passing through the pole. Multiplying by r yields $r^2 = 2r\cos\theta \Rightarrow x^2 + y^2 = 2x \Rightarrow (x-1)^2 + y^2 = 1$, which is a circle of radius $R = 1$ centered at $(1, 0)$. Step 1 · Identify integration bounds: The circle is completely traced as θ goes from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ (or from 0 to π). Step 2 · Set up and evaluate the polar integral.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the area enclosed by the cardioid $r = 1 + \cos\theta$.

Theorem (Area of a Cardioid): The cardioid $r = 1 + \cos\theta$ is symmetric about the polar axis (x-axis) and is completely traced as θ ranges from 0 to 2π. $A = \frac{1}{2}\int_0^{2\pi} [r(\theta)]^2 \, d\theta$. Step 1 · Expand $[r(\theta)]^2$: $(1 + \cos\theta)^2 = 1 + 2\cos\theta + \cos^2\theta$. Step 2 · Apply the trigonometric power-reduction identity $\cos^2\theta = \frac{1 + \cos(2\theta)}{2}$.

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9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Consider the function $f(x)$ defined by its Maclaurin series $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!\,3^{2n+1}}$. Let $g(x) = \int_0^x f(t)dt$. The function $f(x)$ is a known elementary function. Determine the value of $g''(0)$.

The Maclaurin series for $f(x)$ is given by $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!\,3^{2n+1}}$. This is the Maclaurin series for $\sin(u)$ with $u = \frac{x}{3}$. Therefore, $f(x) = \sin\left(\frac{x}{3}\right)$. We are given $g(x) = \int_0^x f(t)dt$. To find $g''(0)$, we first find the derivatives of $g(x)$: First derivative: $g'(x) = \frac{d}{dx}\int_0^x f(t)dt = f(x)$ by the Fundamental Theorem of Calculus. Second derivative: $g''(x) = f'(x)$. We need to evaluate $g''(0)$, which is equal to.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 1 · FoundationsOn an AP Calculus BC exam question, a student writes $\int_0^5 (R(t) - D(t))\,dt = 42.18$ as their final response to a question asking for the total amount of water in a tank at $t = 5$, given initial volume 100 gallons. Identify the missing component in the student's solution.

The student computed only the net change $\int_0^5 (R(t) - D(t))\,dt$, failing to add the initial condition $W(0) = 100$. By the Fundamental Theorem of Calculus: $W(5) = W(0) + \int_0^5 (R(t) - D(t))\,dt = 100 + 42.18 = \mathbf{142.18}$ gallons. Failing to include the initial condition is a common fatal error on FRQ Question 1.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 3 · Advanced / AP FRQLet $y = f(x)$ be the particular solution to $\frac{dy}{dx} = xy^2$ with $f(1) = -1$. Use Euler's method with step size $\Delta x = 0.5$ starting at $x = 1$ to approximate $f(2)$, and determine whether the approximation is an overestimate or underestimate by computing $\frac{d^2y}{dx^2}$ at $(1, -1)$.

Step 1 · Euler Iteration: • $(x_0, y_0) = (1, -1)$. Slope $m_0 = (1)(-1)^2 = 1$. $y_1 = -1 + (1)(0.5) = -0.5$. Point $(1.5, -0.5)$. • $(x_1, y_1) = (1.5, -0.5)$. Slope $m_1 = (1.5)(-0.5)^2 = 1.5(0.25) = 0.375$. $y_2 = -0.5 + (0.375)(0.5) = -0.5 + 0.1875 = \mathbf{-0.3125}$. Step 2 · Second Derivative Evaluation: $\frac{d^2y}{dx^2} = \frac{d}{dx}[xy^2] = (1)y^2 + x(2yy') = y^2 + 2xy(xy^2) = y^2 + 2x^2y^3$. At $(1, -1)$: \[\left. \frac{d^2y}{dx^2} \right|_{(1, -1)} = (-1)^2 + 2(1)^2(-1)^3 = 1 + 2(-1) = 1 - 2 =.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 3 · Advanced / AP FRQLet $f(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$. Identify the closed-form elementary function that this series represents, and determine its radius of convergence.

Write out the terms: $f(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \ldots$ Recall the definition of the hyperbolic sine function: $\sinh(x) = \frac{e^x - e^{-x}}{2}$. Expanding e^x and e^{-x} shows that only odd powers remain with coefficients $1/(2n+1)!$. Thus $f(x) = \sinh\mathbf{(x)}$. Apply the Ratio Test: \[\lim_{n\to\infty} \frac{x^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{x^{2n+1}} = x^2 \lim_{n\to\infty} \frac{1}{(2n+3)(2n+2)} = 0

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13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · MediumOn Section II Part A (Calculator Active), a student is asked for the acceleration vector at $t = 3$ for a particle whose velocity is $\vec{v}(t) = \langle \ln(t^2 + 1), 3\cos(t) \rangle$. What calculator command must be used to find this vector, and what is its 3-decimal numerical value?

Acceleration is the derivative of velocity: $\vec{a}(3) = \langle x''(3), y''(3) \rangle$. Use the calculator's numerical derivative function nDeriv: • $x''(3) = \frac{d}{dt}[\ln(t^2 + 1)]\Big|_{t=3} = \frac{2(3)}{3^2+1} = \frac{6}{10} = 0.6$. • $y''(3) = \frac{d}{dt}[3\cos(t)]\Big|_{t=3} = -3\sin(3) \approx -0.423$. Thus: $\mathbf{\vec{a}(3) = \langle 0.600, -0.423 \rangle}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 2 · MediumLet $\frac{dy}{dx} = x + 2y$ with $y(0) = 1$. Use Euler's method with step size $\Delta x = 0.2$ to approximate $y(0.4)$.

Step 1 (from $x = 0$ to 0.2): $x_0 = 0, y_0 = 1$. Slope $m_0 = 0 + 2(1) = 2$. $y_1 = 1 + (2)(0.2) = 1 + 0.4 = 1.4$. Step 2 (from $x = 0.2$ to 0.4): $x_1 = 0.2, y_1 = 1.4$. Slope $m_1 = 0.2 + 2(1.4) = 0.2 + 2.8 = 3.0$. $y_2 = 1.4 + (3.0)(0.2) = 1.4 + 0.6 = \mathbf{2.0}$. Approximate value $y(0.4) \approx \mathbf{2.0}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests**

[AP Calculus BC Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

Level 1 · FoundationsWhich single unit on the AP Calculus BC exam carries the highest individual weighting, and what specific topic within it is virtually guaranteed to appear on Section II as Free Response Question 6?

Unit 10 (Infinite Sequences and Series) carries the highest individual weighting at 17–18% of the total exam. On Section II Part B, Question 6 is historically and systematically dedicated to Taylor and Maclaurin polynomials, series representations, radius/interval of convergence, and error bounds (Lagrange or Alternating Series Error Bound).

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17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Power Series and Interval of Convergence**

Level 1 · FoundationsA population satisfies $\frac{dP}{dt} = 0.05P - 0.0001P^2$. Determine the carrying capacity L and evaluate $\lim_{t\to\infty} P(t)$ given initial condition $P(0) = 150$.

Factor the differential equation: $\frac{dP}{dt} = 0.0001P(500 - P)$. The carrying capacity is $L = 500$. Since $P_0 = 150 > 0$, as $t \to \infty$, the population approaches its carrying capacity asymptotically: $\lim_{t\to\infty} P(t) = \mathbf{500}$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 Taylor and Maclaurin Series**

[Unit 7BC · Logistic Models & Euler's Method · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Integration by Parts**

Level 2 · MediumDuring Section II Part A (30 minutes, Calculator Active), a student completes Questions 1 and 2 with 5 minutes remaining. When time is called and calculators are stored for Part B (60 minutes, No Calculator), may the student return to Questions 1 and 2 to add work, and what restrictions apply?

Yes. Students are permitted to read, edit, and add work to Questions 1 and 2 at any point during the remaining 60 minutes of Section II Part B. However, they may NOT use a calculator under any circumstances during Part B. Any remaining work on Questions 1 and 2 must be written using analytic notation, non-calculator algebra, or conceptual justifications.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals**

Level 3 · AdvancedA differential equation $\frac{dy}{dx} = (y-1)^2\cos(x)$ has initial condition $y(0) = 2$. Find the particular solution $y = f(x)$.

Separate variables: $\frac{dy}{(y-1)^2} = \cos(x)\,dx$. Integrate both sides: $-\frac{1}{y-1} = \sin(x) + C$. Apply initial condition $x = 0, y = 2$: $-\frac{1}{2-1} = \sin(0) + C \Rightarrow -1 = C$. Substitute $C = -1$:

$-\frac{1}{y-1} = \sin(x) - 1 \Rightarrow \frac{1}{y-1} = 1 - \sin(x) \Rightarrow y - 1 = \frac{1}{1-\sin(x)} \Rightarrow \mathbf{y = 1 + \frac{1}{1-\sin(x)}}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $\frac{dy}{dx} = x + y$ and initial condition $y(0) = 1$, use Euler's method with step size $\Delta x = 0.5$ to approximate $y(1)$.

a. 2.25 ✓

b. 2.00

c. 2.50

d. 2.75

Step 1: at $(0, 1)$, slope $= 0 + 1 = 1$. $\Delta y = 1(0.5) = 0.5 \Rightarrow y(0.5) \approx 1.5$. Step 2: at $(0.5, 1.5)$, slope $= 0.5 + 1.5 = 2$. $\Delta y = 2(0.5) = 1.0 \Rightarrow y(1) \approx 1.5 + 1.0 = 2.25$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $y'' > 0$ (concave up) on the interval, does Euler's method yield an overestimate or underestimate of the true solution?

a. **An underestimate because tangent segments lie below the curve** ✓

b. An overestimate because tangent segments lie above the curve

c. Exact value

d. Depends on the step size

Euler's method uses tangent line segments. When the solution curve is concave up ($y'' > 0$), the curve bends above the tangent segments, making Euler's method an underestimate.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $\frac{dy}{dx} = 2x$ with $y(1) = 3$, what is the error in approximating $y(2)$ using Euler's method with $\Delta x = 0.5$?

a. 0.5 ✓

b. 1.0

c. 0

d. 0.25

Euler: $y(1.5) = 3 + 2(1)(0.5) = 4$. $y(2) = 4 + 2(1.5)(0.5) = 4 + 1.5 = 5.5$. Exact: $y = x^2 + 2 \Rightarrow y(2) = 6$. Error $= 6 - 5.5 = 0.5$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A population grows according to $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$. What is the carrying capacity of this population?

a. 500 ✓

b. 0.04

c. 250

d. $12,500$

In the standard logistic equation $\frac{dP}{dt} = kP(1 - \frac{P}{M})$, the parameter M is the carrying capacity, which is 500.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the logistic model $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$, at what population size is the growth rate $\frac{dP}{dt}$ maximized?

a. 250 ✓

b. 500

c. 125

d. 375

The rate of growth is maximized at the point of inflection of the logistic curve, which occurs at exactly half the carrying capacity: $\frac{M}{2} = \frac{500}{2} = 250$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is $\lim_{t \to \infty} P(t)$ for a logistic population with carrying capacity 800 if $P(0) = 50$?

a. 800 ✓

b. ∞

c. 400

d. 50

For any positive initial population $P_0 > 0$, the logistic curve asymptotically approaches the carrying capacity $M = 800$ as $t \to \infty$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the general form of the solution to the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{M})$?

a. $P(t) = \frac{M}{1 + Ae^{-kt}}$ ✓

b. $P(t) = Me^{kt}$

c. $P(t) = M - Ae^{-kt}$

d. $P(t) = \frac{M}{1 - Ae^{kt}}$

By separation of variables and partial fractions, the logistic solution is $P(t) = \frac{M}{1 + Ae^{-kt}}$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $P(t) = \frac{1000}{1 + 9e^{-0.2t}}$, what is the initial population $P(0)$?

a. 100 ✓

b. 1000

c. 90

d. 10

$P(0) = \frac{1000}{1 + 9(1)} = \frac{1000}{10} = 100.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the concavity of the logistic curve when $P(t) > \frac{M}{2}$?

a. **Concave down ($P''(t) < 0$)** ✓

b. Concave up ($P''(t) > 0$)

c. Zero concavity

d. Undefined

For $P < M/2$, the curve is concave up (accelerating growth). For $P > M/2$, the curve is concave down (decelerating growth).

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the autonomous differential equation $\frac{dy}{dx} = y(y - 3)(y + 2)$, what are all equilibrium solutions?

a. $y = 0, y = 3, y = -2$ ✓

b. $y = 3, y = -2$ only

c. $y = 0$ only

d. $x = 0, x = 3, x = -2$

Equilibrium solutions occur where $\frac{dy}{dx} = 0$, giving horizontal lines $y = 0, y = 3,$ and $y = -2$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For $\frac{dy}{dx} = 2y(1 - \frac{y}{4})$, which equilibrium solution is stable (an attractor)?

a. $y = 4$ ✓

b. $y = 0$

c. Both are stable

d. Neither is stable

Solutions starting near $y = 4$ approach 4 as $x \to \infty$ (since $y' > 0$ for $0 < y < 4$ and $y' < 0$ for $y > 4$), making $y = 4$ a stable attractor.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A population of bacteria is modeled by the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{L})$, where $P(t)$ is the population at time t in hours, k is the growth rate constant, and L is the carrying capacity. It is observed that $\frac{dP}{dt} = 200$ when $P = 100$, and the maximum growth rate occurs when $P = 500$. Find the carrying capacity L of this population.

a. **1000** ✓

b. 750

c. 1200

d. 500

The logistic differential equation is given by $\frac{dP}{dt} = kP(1 - \frac{P}{L})$. The growth rate $\frac{dP}{dt}$ is maximized when the population P is half of the carrying capacity L. We are given that the maximum growth rate occurs when $P = 500$. Therefore, $L/2 = 500$, which implies $L = 1000$. We can also use the other piece of information to verify this. When $P = 100$, $\frac{dP}{dt} = 200$. Substituting $P = 100$ and $L = 1000$ into the differential equation: $200 = k(100)(1 - \frac{100}{1000})$.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the logistic differential equation $\frac{dP}{dt} = kP(M - P)$, where $P(t)$ is the population at time t, k is the growth rate constant, and M is the carrying capacity. If $\frac{dP}{dt} = 0.05P(1000 - P)$, what is the carrying capacity of the population?

a. **1000** ✓

b. 0.05

c. 50

d. 0.00005

The logistic differential equation is given by $\frac{dP}{dt} = kP(M - P)$. In the given equation, $\frac{dP}{dt} = 0.05P(1000 - P)$. By comparing the two forms, we can identify the carrying capacity M. We see that $k = 0.05$ and $M = 1000$. Therefore, the carrying capacity of the population is 1000.

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Logistic growth model is $\frac{dP}{dt} =$

a. kP

b. $kP(1 - P/M)$ ✓

c. $k(M - P)$

d. P^2

Logistic: $kP(1 - P/M)$ where M is carrying capacity.

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35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A colony grows according to $\frac{dP}{dt} = 0.11P\left(1 - \frac{P}{1100}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. **M=1100,\ P=550 ✓**

b. M=1100,\ P=1100

c. M=550,\ P=1100

d. M=2200,\ P=550

Carrying capacity $M = 1100$; growth rate is maximal at the inflection point $P = M/2 = 550$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{2000}\right)$, find the instantaneous growth rate when $P = 1000$.

a. -25

b. 25.5

c. **25 ✓**

d. 50

Substitute $P = 1000$: $0.05 \times 1000 \times \left(1 - \frac{1000}{2000}\right) = 25$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider a region R in the xy-plane bounded by the graphs of $y = \frac{1}{2}x^2$ and $y = 2x$. A solid has its base in region R, and for each value of x in the interval $[0, 4]$, the cross-section perpendicular to the x-axis is an equilateral triangle with side length $s(x) = 2x - \frac{1}{2}x^2$. Write an integral expression that represents the volume of this solid. You do not need to evaluate the integral.

a. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right)^2 dx$ ✓

b. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right) dx$

c. $V = \int_0^4 \frac{1}{2}\left(2x - \frac{1}{2}x^2\right)^2 dx$

d. $V = \int_0^4 \frac{\sqrt{3}}{2}\left(2x - \frac{1}{2}x^2\right) dx$

The base of the solid is defined by the region R bounded by $y = \frac{1}{2}x^2$ and $y = 2x$. The upper bound is $y = 2x$ and the lower bound is $y = \frac{1}{2}x^2$. The intersection points are found by setting $\frac{1}{2}x^2 = 2x$, which gives $x^2 = 4x$, so $x^2 - 4x = 0$, and $x(x - 4) = 0$. Thus, the region extends from $x = 0$ to $x = 4$. The side length of the equilateral triangle cross-section is given by $s(x) = 2x - \frac{1}{2}x^2$. The area of an equilateral triangle with side length s is.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 Taylor and Maclaurin Series

An AP Calculus BC FRQ models a population $P(t)$ satisfying the logistic differential equation $\frac{dP}{dt} = \frac{3}{600}P(600 - P)$. At what population size P is the population growing most rapidly?

a. $P = 300$ ✓

b. $P = 600$

c. $P = 150$

d. $P = 0$

In a logistic growth model dP/dt = kP(M - P), the maximum rate of growth occurs at the inflection point where P = M / 2 = 600 / 2 = 300.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Integration by Parts

Consider the function $f(x)$ whose derivative is given by $f'(x) = \frac{\sin(x)}{x^2 + 1}$. A related function $g(x)$ is defined by $g(x) = \int_0^x f(t)dt$. If $f(0) = k$, where k is a positive constant, find the value of $g'(0)$ and determine if $g(x)$ is increasing or decreasing at $x = 0$.

a. $g'(0) = k$**, and** $g(x)$ **is increasing.** ✓

b. $g'(0) = 0$, and $g(x)$ is neither increasing nor decreasing.

c. $g'(0) = 1$, and $g(x)$ is increasing.

d. $g'(0) = k$, and $g(x)$ is decreasing.

We are given $g(x) = \int_0^x f(t)dt$. By the Fundamental Theorem of Calculus Part 1, the derivative of $g(x)$ is $g'(x) = f(x)$. We need to find $g'(0)$. Substituting $x = 0$ into $g'(x)$, we get $g'(0) = f(0)$. We are given that $f(0) = k$, where k is a positive constant. Therefore, $g'(0) = k$. To determine if $g(x)$ is increasing or decreasing at $x = 0$, we examine the sign of $g'(0)$. Since k is a positive constant, $g'(0) = k > 0$. A function is increasing at a point if its derivative at that point is posit.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

For $\frac{dy}{dx} = y,\ y(0) = 1$ (exact solution $y = e^x$), a single Euler step with $h = 0.25$ gives y_1. Compared with the exact value $y(0.25) = e^{0.25}$, Euler's approximation:

a. overestimates the true value

b. equals the true value exactly

c. cannot be compared

d. underestimates the true value ✓

Euler gives $y_1 = 1.25$; true value $e^{0.25} \approx 1.2840$. Since $y = e^x$ is concave up, the tangent-line step underestimates the curve.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Parametric Equations

On an AP Calculus BC Free-Response Question, a particle moves along a curve with velocity vector $\vec{v}(t) = \langle 2t, 3t^2 \rangle$. What is the speed of the particle at time $t = 1$?

a. $\sqrt{13}$ ✓

b. 5

c. $\sqrt{5}$

d. 13

Speed is the magnitude of the velocity vector: ||v(1)|| = sqrt((2*1)^2 + (3*1^2)^2) = sqrt(13).

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.02 Vector-Valued Functions

In a Taylor series BC FRQ, the alternating series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^3}$ is approximated by its first 2 terms S_2. According to the Alternating Series Error Bound, what is the maximum possible error $|S - S_2|$?

a. $\frac{1}{3^3} = \frac{1}{27}$ ✓

b. $\frac{1}{2^3} = \frac{1}{8}$

c. $\frac{1}{3^2} = \frac{1}{9}$

d. $\frac{1}{2 \cdot 3^3}$

By the Alternating Series Error Bound, |S - S_n| <= a_{n+1} = 1 / (2+1)^3 = 1 / 3^3 = 1 / 27.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Polar Curves and Derivatives

For $\frac{dy}{dx} = y - x,\ y(0) = 1$, use Euler's method with step $h = 0.45$ to approximate y after two steps (at $x = 0.9$).

a. 1.45

b. 2.35

c. **1.9** ✓

d. 1.95

$y_1 = y_0 + h(y_0 - x_0) = 1 + 0.45(1 - 0) = 1.45$. $y_2 = y_1 + h(y_1 - x_1) = 1.45 + 0.45(1.45 - 0.45) = 1.9$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.02 Area in Polar Coordinates

For $\frac{dP}{dt} = 0.4P\left(1 - \frac{P}{300}\right),\ P(0) = 60$, use one Euler step with $h = 1$ to approximate $P(1)$.

a. 158.4

b. **79.2** ✓

c. 60

d. 84

$f(0, 60) = 0.4 \times 60 \times (1 - 60/300) = 19.2$. $P_1 = P_0 + hf(x_0, P_0) = 60 + 1(19.2) = 79.2$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

For $\frac{dy}{dx} = x + y,\ y(0) = 0$, use Euler's method with step $h = 0.1$ to approximate y after four steps (at $x = 0.4$).

a. **0.0641** ✓

b. -0.0359

c. 0.1641

d. 0.031

Stepping four times with $h = 0.1$: $y_1 = 0$, $y_2 = 0.01$, $y_3 = 0.031$, $y_4 = 0.0641$.

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